

Digit Test Process

Laptops have to be functionally competent besides being really small, portable and easy to use. To evaluate such purposeful devices purely on the basis of performance seems unfair—we can't be blind to the fact that laptops are severely handicapped in terms of space and power. Hence, we have accorded more weightage to features as well as the ability to add more peripherals. When choosing a laptop, you should first determine the features that are important to you.

1. Performance: We ran office productivity-related benchmarks like Business Winstone 2001 and Content Creation Winstone 2001, besides the BatteryMark 4.01 which tests the battery output. To test for OpenGL and Direct3D compliance, we ran *Quake III Arena* and 3DMARK 2000 to determine whether the laptop was capable of decent graphical output or not (remember no benchmarks were run).

2. Core configuration: The core configuration should be well matched with the critical compo-

nents (CPU, Chipset, RAM and the drive space offered). We set a minimum of 500 MHz of processing power, 64 MB of RAM, and 10 GB of hard drive space as the necessary requirements. This is the least one can expect for running the latest Operating Systems and applications.

3. I/O functionality expandability: Without this the notebook would become an isolated computing device with no ability to communicate with others. These come in the form of serial, parallel, USB, and PCMCIA ports. Our list of minimum required ports is: one Type-II or Type-I PCMCIA slot, one serial port, one parallel port, one PS/2 port, one Mic In / One audio output, two USB ports, one RJ-11 modem jack, one RJ-45 LAN jack, one S-Video out, one 4

Mbps IrDA Comm port.

4. Package contents and accessories: Here we noted the software bundle and package contents of the notebook.

5. Build quality and ergonomics: This is the most subjective part of the test; hence opinions from three individuals were taken. Extra points were awarded for exceptional looks and design.

6. Software bundling: We focused on the operating system here: higher marks were awarded to Notebooks that bundled the Windows XP home edition. Also, more marks were awarded to notebooks that offered office suites, management tools, and some sort of system recovery software.

7. Display: This was checked for the maximum possible resolution, screen size, and the number of inactive pixels, if any.

8. Extra goodies: Extra points were awarded to laptops bundling a lot of extras—DVD, CD-RW, Combo, Superdrive, extra external storage devices, MP3 player capabilities, USB 2.0 functionality, and so on.

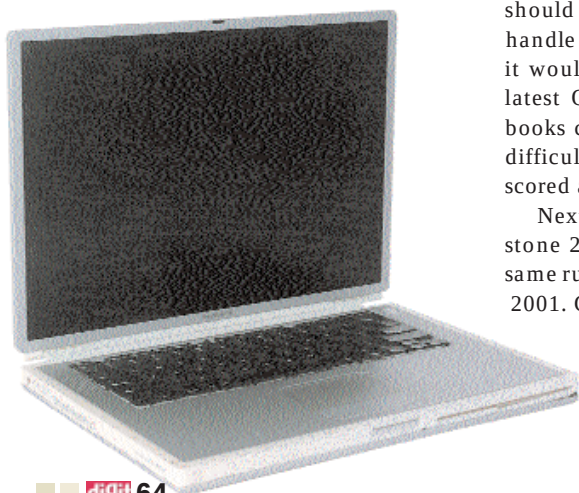


“Convenience, mobility and stamina in terms of battery life takes precedence over performance”

Hatim Kantawalla,
Deputy Head, Test Centre, Jasubhai Digital Media

than the regular 3.5-inch form factor. This also adversely affects the maximum capacity available: to have a 40 GB or higher drive in a laptop is a really tough achievement. The power requirements have also been modified—this makes the drives run slower, averaging a seek time of 13 ms, and only 4,200 or so rpm.

All the notebooks came with at least 20 GB drive space. Compaq Presario 1725 stood tall by providing a whopping 40 GB drive. The ACI Profile 3270 and the ACI Xtreme were not too far behind with 30 GB drives.



Performance

What really matters in day-to-day tasks is the core configuration. Let's take a look at the tests.

The scores for Business Winstone 2001 are important: it is a system-level, application-based benchmark that measures overall performance when running a suite of office productivity applications. Here we sought a range: scores of 20 and below were considered average, between 20 to 30 were adjudged good, and 30 and above were clearly excellent.

Anything that logs a score near 20 should be considered a notebook that can handle most office applications with ease; it wouldn't have problems running the latest OSes. Here we found most notebooks cleared the 20 mark without any difficulty. The Compaq Presario 1725 scored an amazing 53.4.

Next, we ran Content Creation Winstone 2001. Here too you can apply the same rule for scores as Business Winstone 2001. Content Creation is a more multimedia-heavy benchmark and the scores are generally expected to be low. Despite this, we discovered that most of the note-

books could clear the 20 figure with relative ease. The Presario 1725 was no surprise, as it again scored high (51.3). The ACI Profile 3270 and the ACI Xtreme fared well too, scoring 37.4 and 36 respectively. All said and done, the Compaq Presario 1725 showed its pedigree by truly stamping its authority all through the benchmarks. It is a notebook power users would love to possess (and display).

While considering a new laptop, you should be aware of enjoying a computing experience with better display screens and graphics cards. All major laptop manufacturers are integrating formidable graphics cards with full OpenGL and Direct3D functionality, coupled with hardware-based DVD decoding capabilities.

To test the ability of the graphics card to run OpenGL and Direct 3D games and applications, we ran 3DMARK 2000 and *Quake III: Arena* as part of a passive test (passive because the test was only to judge whether the notebook was able to run the application). Surprisingly, we found that most notebooks could actually run *Quake III*—and some could run it pretty well! As a matter of fact, the ACI Xtreme came out as the top dog: it blazed away in *Quake III*